IGCSE 0607 Practice Questions — E1.12 — Combined Questions & Markscheme

Percentages

Total Questions: 9 — **Total Marks:** 52 Marks

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1 **(a)** Write 24 as a percentage of 60.

..... % **[2]**

(b) A coat has a normal price of \$80. In a sale, the price is reduced by 15%.
Work out the sale price of the coat.

\$ ..... **[2]**

2 A shop sells a jacket for \$120. During a sale, the shop offers a 15% discount on the price of the jacket.

(a) Calculate the sale price of the jacket.

\$ ..... **[2]**

(b) The original price of \$120 gives the shop a profit of 20% on the cost price.
Calculate the cost price of the jacket.

\$ ..... **[3]**

3 **(a)** In 2021, the population of a town was 45 000.
In 2022, the population increased by 4.2%.
Calculate the population in 2022.

..... **[2]**

(b) In a sale, the price of a television is reduced by 24%.
The sale price is \$342.
Calculate the original price of the television.

\$ ..... **[3]**

4 Amira invests \$4500 in a savings account that pays compound interest at a rate of 2.4% per year.

(a) Calculate the total amount in Amira's account at the end of 5 years.

\$ ..... **[3]**

(b) Ben invests \$4500 in a different savings account that pays simple interest at a rate of r% per year. At the end of 5 years, Ben has the same total amount in his account as Amira has in hers.

Calculate the value of r.

$r =$ ..... **[3]**

5 The population of a city increases by $x\%$ each year.
In 2015, the population was 1 250 000.
In 2018, the population was 1 367 631.

(a) Show that the value of x is approximately 3.

[3]

(b) Assume the population continues to increase by exactly 3% each year.
Calculate the year in which the population will first exceed double its 2015 value.

..... **[4]**

6 **(a)** Ali invests \$5000 at a rate of 2.5% per year compound interest. Calculate the value of his investment at the end of 3 years.
Give your answer correct to the nearest cent.

\$ ..... **[2]**

(b) Beatrice invests a sum of money at a rate of r% per year compound interest.
After 5 years, the value of her investment has increased by 30%.
Calculate the value of r.

$r =$ ..... **[3]**

(c) The value of a car depreciates by 12% each year.
After 3 years, the value of the car is \$17 036.80.
Calculate the original value of the car.

\$ ..... **[3]**

7 A shopkeeper buys a television for \$320. He sells it for \$400.
Calculate his percentage profit.

..... % **[2]**

8 **(a)** Amira invests \$4500 in a savings account paying compound interest at a rate of 2.4% per year.
Calculate the total value of her investment at the end of 5 years.

\$ ..... **[3]**

(b) In a sale, the price of a computer is reduced by 18%.
The sale price is \$557.60.
Calculate the original price of the computer.

\$ ..... **[3]**

9 **(a)** A car's value depreciates by 15% in the first year.
In each subsequent year, its value depreciates by 8%.
After exactly 3 years, the value of the car is \$17 986.
Calculate the original value of the car.

\$ ..... **[4]**

(b) Calculate the overall percentage decrease in the car's value over the 3 years.
Give your answer correct to 1 decimal place.

..... % **[2]**

(c) The population of a city is P.
The population increases by r% each year.
After 12 years, the population has exactly doubled.
Calculate the value of r.

$r =$ ..... **[3]**

1 **(a)** $\frac{24}{60} \times 100$ **[M1]**

40 **[A1]**

(b) $\frac{15}{100} \times 80 = 12$ OR 80×0.85 **[M1]**

68 **[A1]**

[Back to Question] *[Total Marks: 4]*

2 **(a)** Discount $= 0.15 \times 120 = 18$ OR 0.85×120 **[M1]**

102 **[A1]**

(b) Let cost price be C. $1.2C = 120$ **[M1]**

$C = \frac{120}{1.2}$ **[M1]**

100 **[A1]**

[Back to Question] *[Total Marks: 5]*

3 **(a)** $45\,000 \times 1.042$ OR $45\,000 + 45\,000 \times 0.042$ **[M1]**

46 890 **[A1]**

(b) $100\% - 24\% = 76\%$ **[B1]**

$\frac{342}{0.76}$ OR $\frac{342}{76} \times 100$ **[M1]**

450 **[A1]**

[Back to Question] *[Total Marks: 5]*

4 **(a)** $4500 \times (1 + \frac{2.4}{100})^5$ or 4500×1.024^5 **[M2]**

If M0, M1 for 4500×1.024 or 4608 seen.

5066.57 or 5066.56 . . . **[A1]**

(b) Interest earned $= 5066.57 - 4500 = 566.57$ (accept unrounded) **[M1]**

$\frac{4500 \times r \times 5}{100} = 566.57$ OR $225r = 566.57$ **[M1]**

2.52 or 2.518 . . . **[A1]**

[Back to Question] *[Total Marks: 6]*

5 **(a)** $1\,250\,000 \times (1 + \frac{x}{100})^3 = 1\,367\,631$ **[M1]**

$(1 + \frac{x}{100})^3 = 1.0941048$ **[M1]**

$1 + \frac{x}{100} = \sqrt[3]{1.0941048} = 1.0304\cdots \implies x = 3.04\cdots \approx 3$ **[A1]**

(b) Target population $= 2 \times 1\,250\,000 = 2\,500\,000$ **[B1]**

$1\,250\,000 \times (1.03)^n > 2\,500\,000 \implies 1.03^n > 2$ **[M1]**

Using logs: $n\log(1.03) > \log(2)$ OR trial and improvement yielding n between 23 and 24 **[M1]**

$n > 23.449\ldots$, so it takes 24 years. Year $= 2015 + 24 = 2039$ **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** $5000 \times \left(1 + \frac{2.5}{100}\right)^3$ **[M1]**

5384.45 **[A1]**

(b) $\left(1 + \frac{r}{100}\right)^5 = 1.30$ **[M1]**

$1 + \frac{r}{100} = \sqrt[5]{1.30}$ **[M1]**

5.39 (or $5.387\ldots$) **[A1]**

(c) $V \times (1 - 0.12)^3 = 17\,036.80$ **[M1]**

$V = \frac{17\,036.80}{0.88^3}$ **[M1]**

25 000 **[A1]**

[Back to Question] *[Total Marks: 8]*

7 Profit $= \$400 - \$320 = \$80$ or $\frac{400}{320} \times 100$

[M1]

25 **[A1]**

[Back to Question] *[Total Marks: 2]*

8 **(a)** $4500 \times \left(1 + \frac{2.4}{100}\right)^5$ or 4500×1.024^5 **[M2]**

[M1] for 4500×1.024^n where $n > 1$

5066.55 **[A1]**

(b) $100\% - 18\% = 82\%$ or 0.82 **[B1]**

$\frac{557.60}{0.82}$ or $x \times 0.82 = 557.60$ **[M1]**

680 **[A1]**

[Back to Question] *[Total Marks: 6]*

9 **(a)** Expression for 3 years depreciation: e.g. $V \times (1 - 0.15) \times (1 - 0.08)^2$ or $V \times 0.85 \times 0.92^2$ **[M2]**

[M1] for $V \times 0.85$ or $V \times 0.92^2$ seen

$V \times 0.71944 = 17986$ or $V = \dfrac{17986}{0.85 \times 0.92^2}$ **[M1]**

25 000 **[A1]**

(b) $\dfrac{25000 - 17986}{25000} \times 100$ or $100\% - (0.85 \times 0.92^2 \times 100)\%$ **[M1]**

28.1 **[A1]**

(c) $P \times \left(1 + \dfrac{r}{100}\right)^{12} = 2P$ or $\left(1 + \dfrac{r}{100}\right)^{12} = 2$ **[M1]**

$1 + \dfrac{r}{100} = \sqrt[12]{2}$ or 1.059... **[M1]**

5.95 (correct to 3 significant figures) **[A1]**

[Back to Question] *[Total Marks: 9]*